

2017-Ph-H1-Q5

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

Two masses are connected by a light string over a pulley. What is the acceleration?

Assume:

- $m_1 > m_2$
- No friction
- Use Newton's second law

 **Apply Newton's Laws:**

Total mass: $m_1 + m_2$

Net force: $(m_1 - m_2)g$

So:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$

 **Final Answer: A** — $\frac{(m_1 - m_2)g}{m_1 + m_2}$

 **Revision Tips:**

- Use $F = ma$ for the **whole system**
 - Assume tension is internal to system
 - Works only for **frictionless pulleys**
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